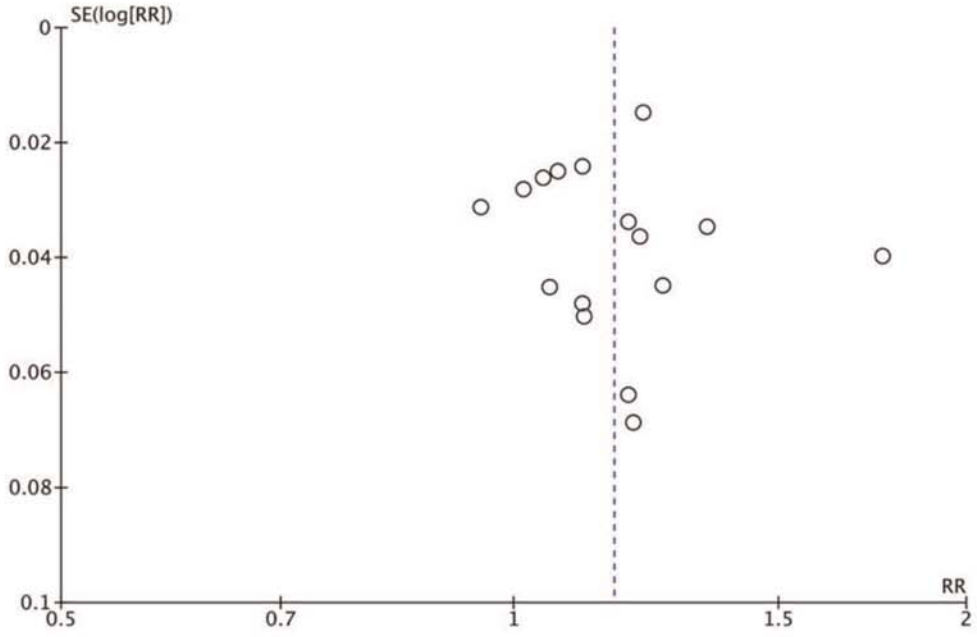
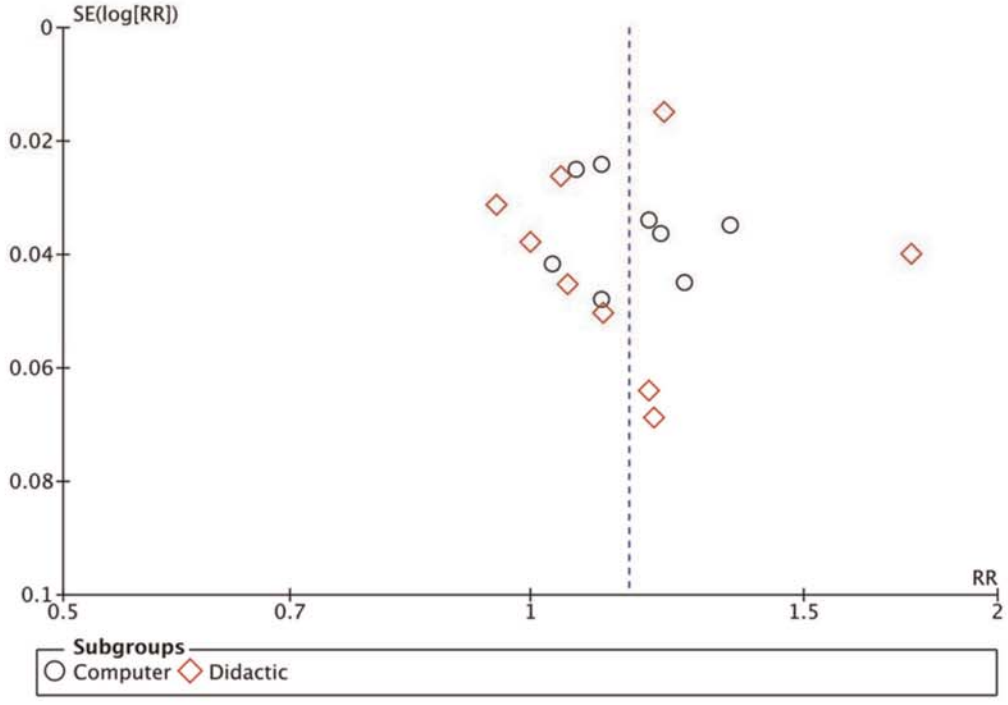
Supplementary material

Supplementary Fig. 1 Funnel plots for assessing the impact of training on optical diagnosis accuracy of small/diminutive colorectal polyps in **A** all studies and **B** computer-based and didactic training in all endoscopists

A

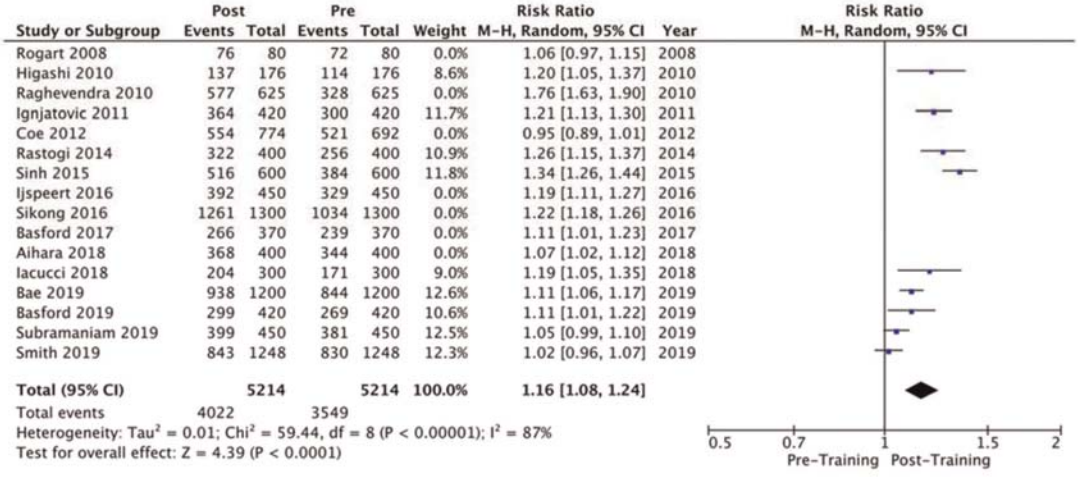


B



Supplementary material

Supplementary Fig. 2 Forest plots for all studies assessing the effect of training on accuracy of optical diagnosis of small/diminutive colorectal polyps only



Study or Subgroup	Post Events	Post Total	Pre Events	Pre Total	Weight	Risk Ratio M-H, Random, 95% CI	Year
Rogart 2008	76	80	72	80	0.0%	1.06 [0.97, 1.15]	2008
Higashi 2010	137	176	114	176	8.6%	1.20 [1.05, 1.37]	2010
Raghevendra 2010	577	625	328	625	0.0%	1.76 [1.63, 1.90]	2010
Ignjatovic 2011	364	420	300	420	11.7%	1.21 [1.13, 1.30]	2011
Coe 2012	554	774	521	692	0.0%	0.95 [0.89, 1.01]	2012
Rastogi 2014	322	400	256	400	10.9%	1.26 [1.15, 1.37]	2014
Sinh 2015	516	600	384	600	11.8%	1.34 [1.26, 1.44]	2015
Ijspeert 2016	392	450	329	450	0.0%	1.19 [1.11, 1.27]	2016
Sikong 2016	1261	1300	1034	1300	0.0%	1.22 [1.18, 1.26]	2016
Basford 2017	266	370	239	370	0.0%	1.11 [1.01, 1.23]	2017
Aihara 2018	368	400	344	400	0.0%	1.07 [1.02, 1.12]	2018
Iacucci 2018	204	300	171	300	9.0%	1.19 [1.05, 1.35]	2018
Bae 2019	938	1200	844	1200	12.6%	1.11 [1.06, 1.17]	2019
Basford 2019	299	420	269	420	10.6%	1.11 [1.01, 1.22]	2019
Subramaniam 2019	399	450	381	450	12.5%	1.05 [0.99, 1.10]	2019
Smith 2019	843	1248	830	1248	12.3%	1.02 [0.96, 1.07]	2019
Total (95% CI)		**5214**		**5214**	**100.0%**	**1.16 [1.08, 1.24]**	
Total events	4022		3549				

Heterogeneity: $Tau^2 = 0.01$; $Chi^2 = 59.44$, df = 8 ($P < 0.00001$); $I^2 = 87\%$
Test for overall effect: $Z = 4.39$ ($P < 0.0001$)

Appendix 1 a PRISMA checklist. (From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097)

Supplementary material

	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	4-5
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	6-7
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	8-9
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	8
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	8-9
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	8
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	8
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	8-9
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	9
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8-9

Supplementary material

Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	9-10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	10-11
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	10-11

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	9-10
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	10-11
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	12
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	12, Table 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	15
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	12-15, Figures 2-4 Supplementary figure 1-2
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	12-15
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	15
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	12-15
DISCUSSION			

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Supplementary material

Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	16-19
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17-18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	18-19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	2

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: **www.prisma-statement.org**.

Thieme

Supplementary material

Appendix 1 b MOOSE checklist. (From: Stroup DF, Berlin JA, Morton SC et al. for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. JAMA. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008)

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	1-2
2	Hypothesis statement	1-2
3	Description of study outcome(s)	10
4	Type of exposure or intervention used	10
5	Type of study designs used	9
6	Study population	9
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	9
8	Search strategy, including time period included in the synthesis and key words	8
9	Effort to include all available studies, including contact with authors	9
10	Databases and registries searched	8
11	Search software used, name and version, including special features used (eg, explosion)	8
12	Use of hand searching (eg, reference lists of obtained articles)	8
13	List of citations located and those excluded, including justification	Figure 1
14	Method of addressing articles published in languages other than English	9
15	Method of handling abstracts and unpublished studies	9
16	Description of any contact with authors	9
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	9
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	9
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	9
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	N/A
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	9
22	Assessment of heterogeneity	10-11
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	10-11

Supplementary material

24	Provision of appropriate tables and graphics	Table 1 and Figure 2-4, Supplementary material
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	Figure 2-4
26	Table giving descriptive information for each study included	Table 1
Item No	**Recommendation**	**Reported on Page No**
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	15
30	Justification for exclusion (eg, exclusion of non-English language citations)	8-9
31	Assessment of quality of included studies	15
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	17
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	17-18
34	Guidelines for future research	18-19
35	Disclosure of funding source	2

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.